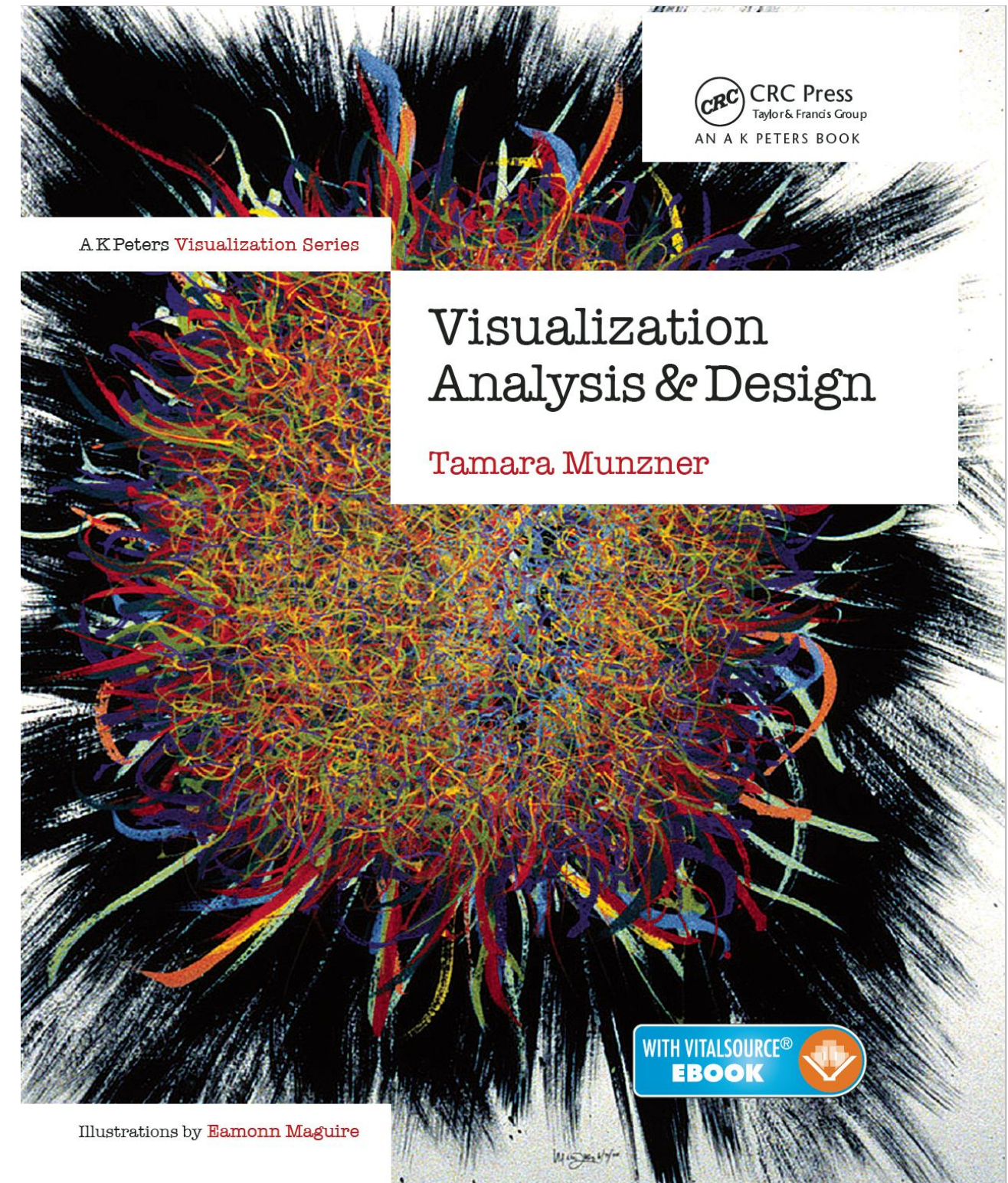


CDS505

Data Visualization & Visual Analytics

Reducing Items and Attributes

*Nur Intan Raihana Ruhaiyem, PhD.
School of Computer Sciences
Universiti Sains Malaysia*



- ✓ Introduction
- ✓ Filtering
- ✓ Item Aggregation
- ✓ Attribute Aggregation: Dimensionality Reduction

Reducing Items and Attributes

- Reduce/increase: inverses (bi-directional)
- **Filter**
 - pro: straightforward and intuitive to understand and compute
 - con: out of sight, out of mind
- **Aggregation**
 - pro: inform about whole set
 - con: difficult to avoid losing signal
- **Not mutually exclusive**
 - combine filter, aggregate
 - combine reduce, change, facet

Reducing Items and Attributes

➔ Filter

→ Items



→ Attributes

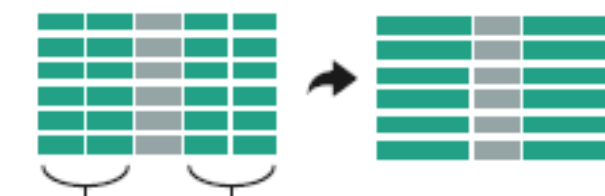


➔ Aggregate

→ Items



→ Attributes



Reduce

➔ Filter



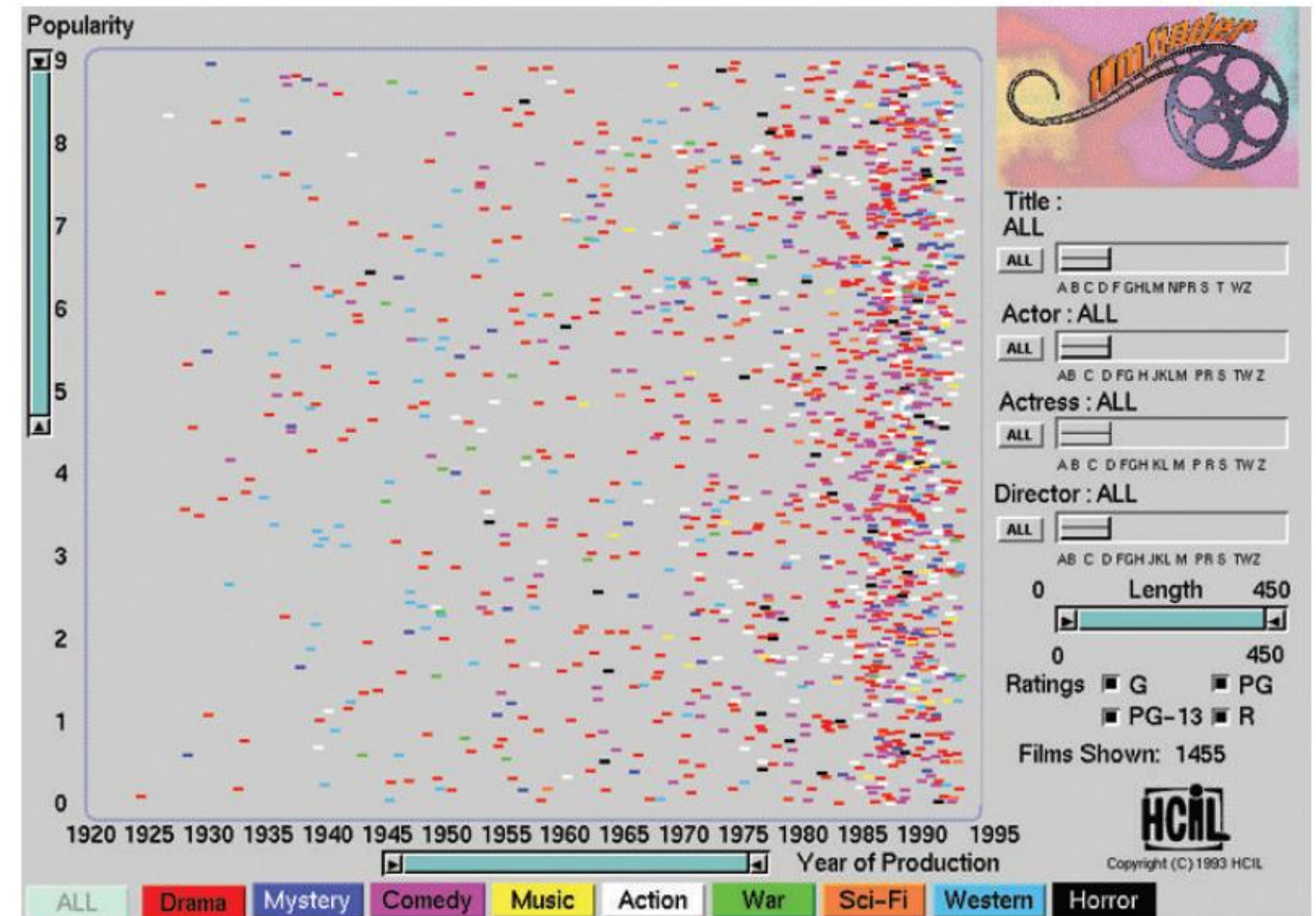
➔ Aggregate



➔ Embed



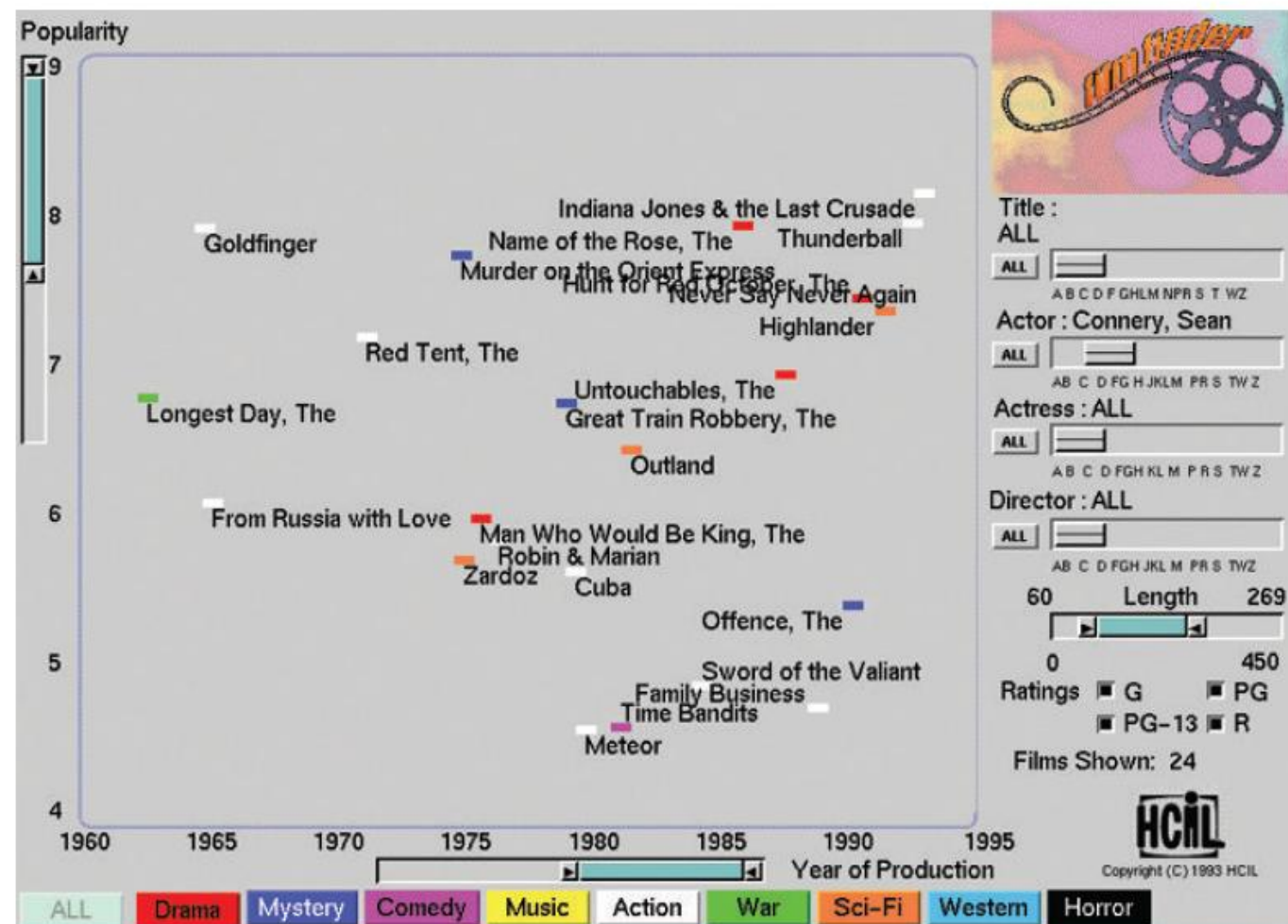
- **Item filtering** - browse through tightly coupled interaction, alternative to queries that might return far too many or too few
 - **Dynamic Filtering/Queries** – **FilmFinder** features tightly coupled **interactive filtering**, where the result of moving sliders and pressing buttons is immediately reflected in the visual encoding.
- a) Exploration begins with an overview of all movies in the dataset.



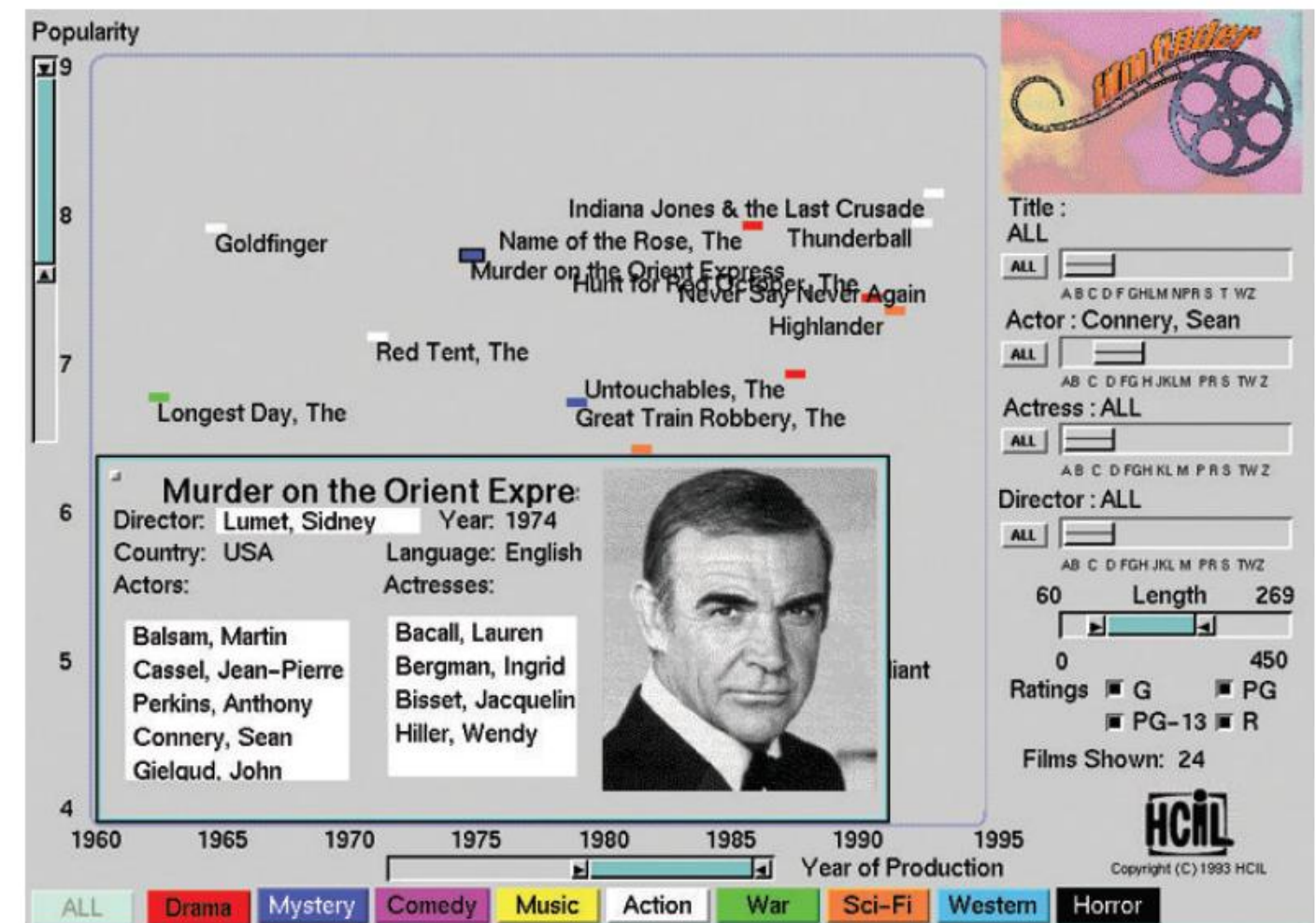
(a)

Filtering

(b) Moving the actor slider to select Sean Connery filters out most of the other movies, leaving enough room to draw labels. (c) Clicking on the mark representing a movie brings up a detail view. From [Ahlberg and Shneiderman 94].



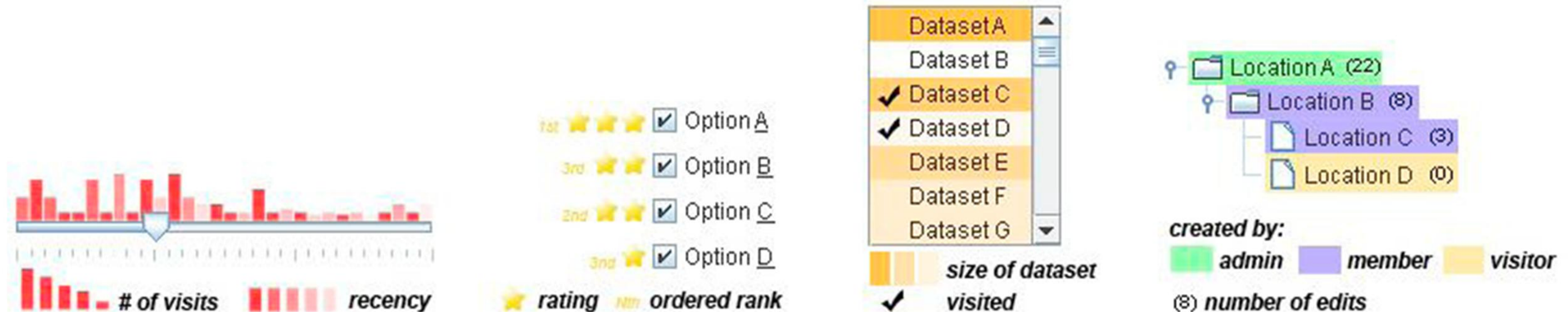
(b)



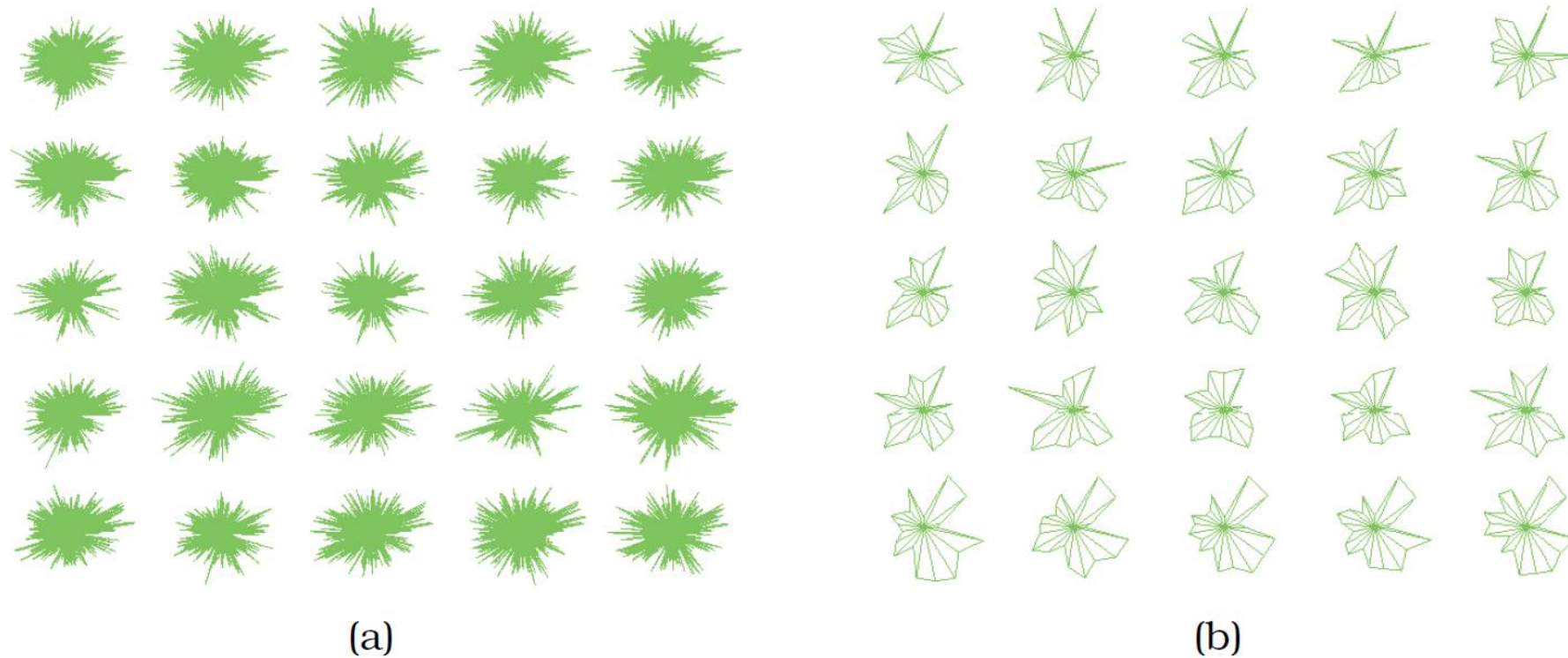
(c)

Scented widgets

- augment widgets for filtering to show *information scent*
 - cues to show whether value in drilling down further vs looking elsewhere
- concise, in part of screen normally considered control panel



The scented widget idiom adds visual encoding information directly to standard graphical widgets to make filtering possible with high information density displays. From [Willett et al. 07, Figure 2].

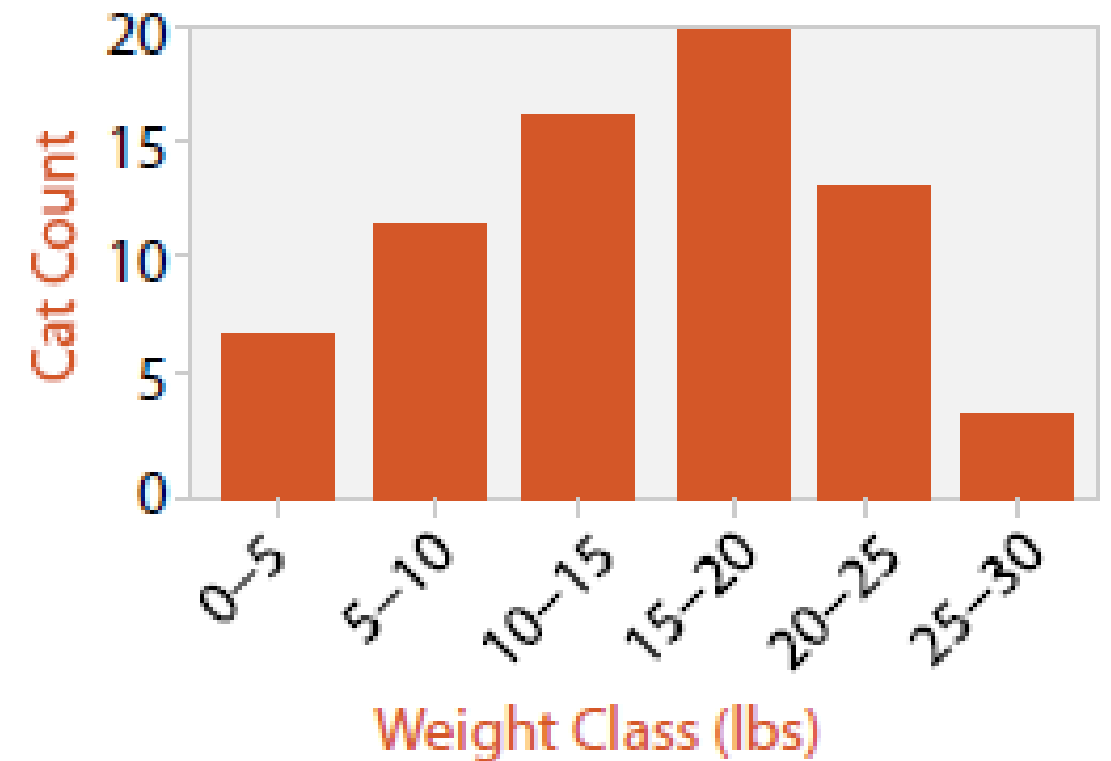


Attribute filtering

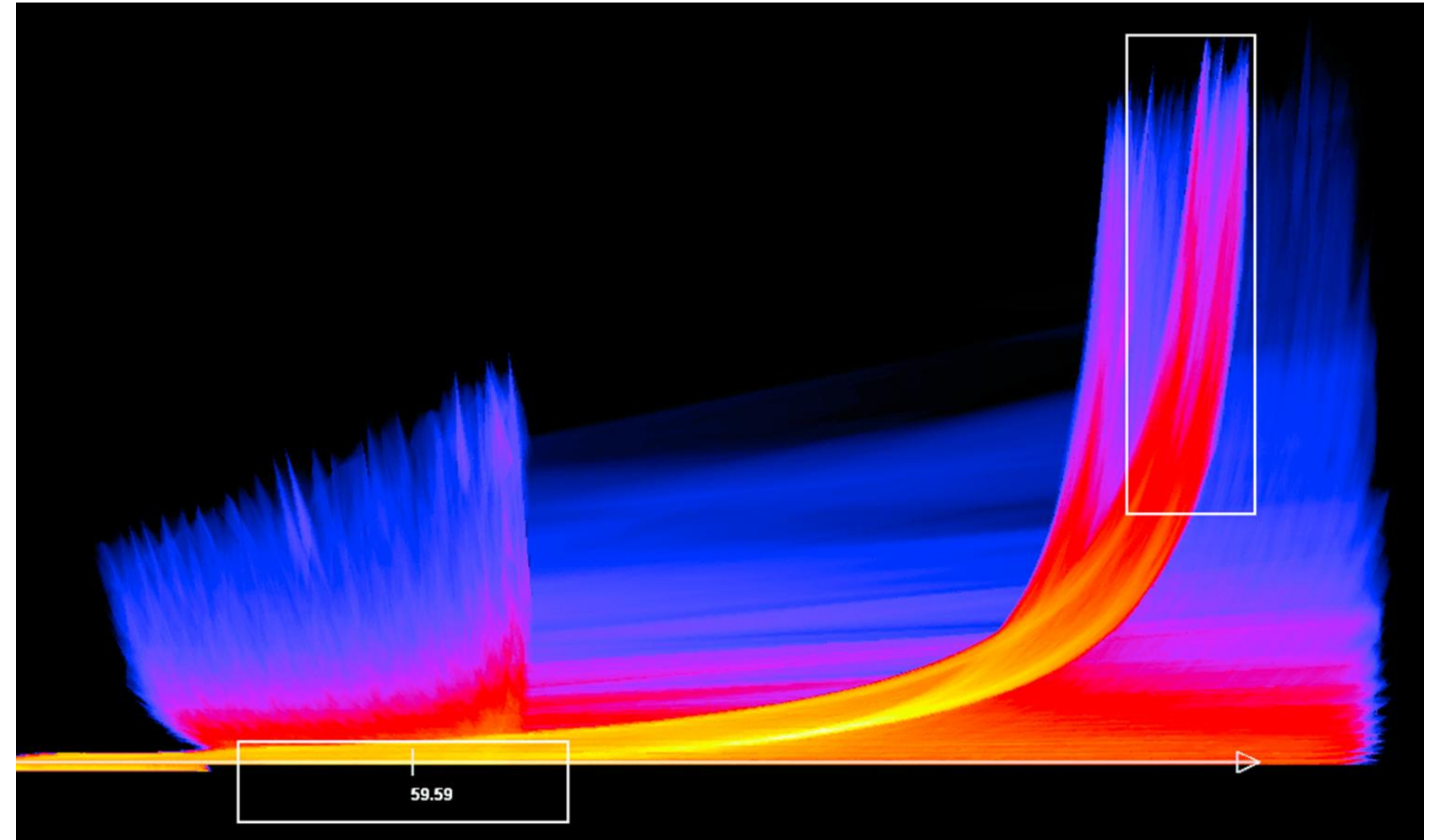
- **DOSFA** (Dimensional Ordering, Spacing, and Filtering Approach) - shown on star glyphs with 215 dimensions and 298 points. (a) The full dataset is so dense that patterns cannot be seen. (b) After ordering on similarity and filtering on both similarity and importance, the star glyphs show structure. From [Yang et al. 03].
- Attribute filtering is often used in conjunction with **attribute ordering**.

- So that a group of elements is represented by a new **derived element** that stands in for the entire group.
- Elements are merged together with aggregation, as opposed to eliminate completely with filtering.
- Aggregation and filtering can be used in conjunction with each other.
- As with filtering, aggregation can be used for both **items** and **attributes**.
- Aggregation typically involves the use of a derived attribute. A very simple example is computing an average; the four other basic aggregation operators are minimum, maximum, count, and sum.

- **Idiom: Histogram** - aggregates an arbitrary number of items into a concise representation of their distribution.
 - **Static** item aggregation
 - Task: find distribution
 - Data: table
 - Derived data
 - new table: keys are bins, values are counts
 - Bin size crucial
 - pattern can change dramatically depending on discretization
 - opportunity for interaction: control bin size on the fly



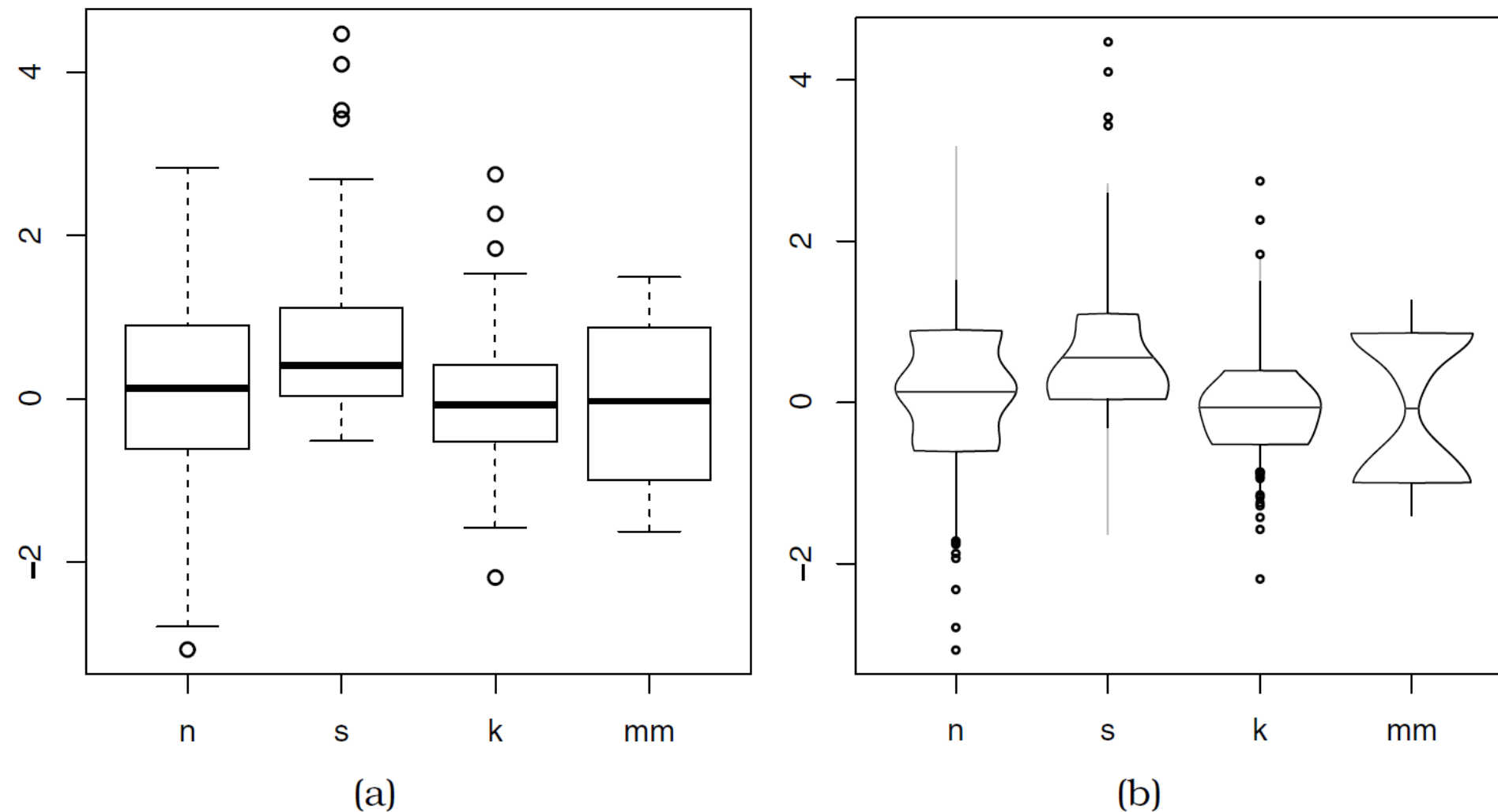
- **Idiom: Continuous scatterplot**
 - Static item aggregation
 - Data: table
 - Derived data: table (overplot density instead of plotting individual point)
 - Key attribs: x,y for pixels, Quant attrib: overplot density
 - Dense space-filling 2D matrix
 - Colour: sequential categorical hue + ordered luminance colourmap



Tornado air-flow direction – magnitude (horizontal) vs z-direction velocity (Vertical), shows the density at each location, solving the problem of occlusion from overplotting and allowing scalability to large datasets. From [Bachthaler and Weiskopf 08, Figure 9].

- **Idiom: Boxplot** - static item aggregation
 - Task: find distribution
 - Data: table
 - Derived data
 - 5 quant attributes:
 - **median**: central line
 - **lower and upper quartile**: boxes
 - **lower and upper fences**: whiskers -values beyond which items are outliers
 - outliers beyond fence cutoffs explicitly shown

Item Aggregation



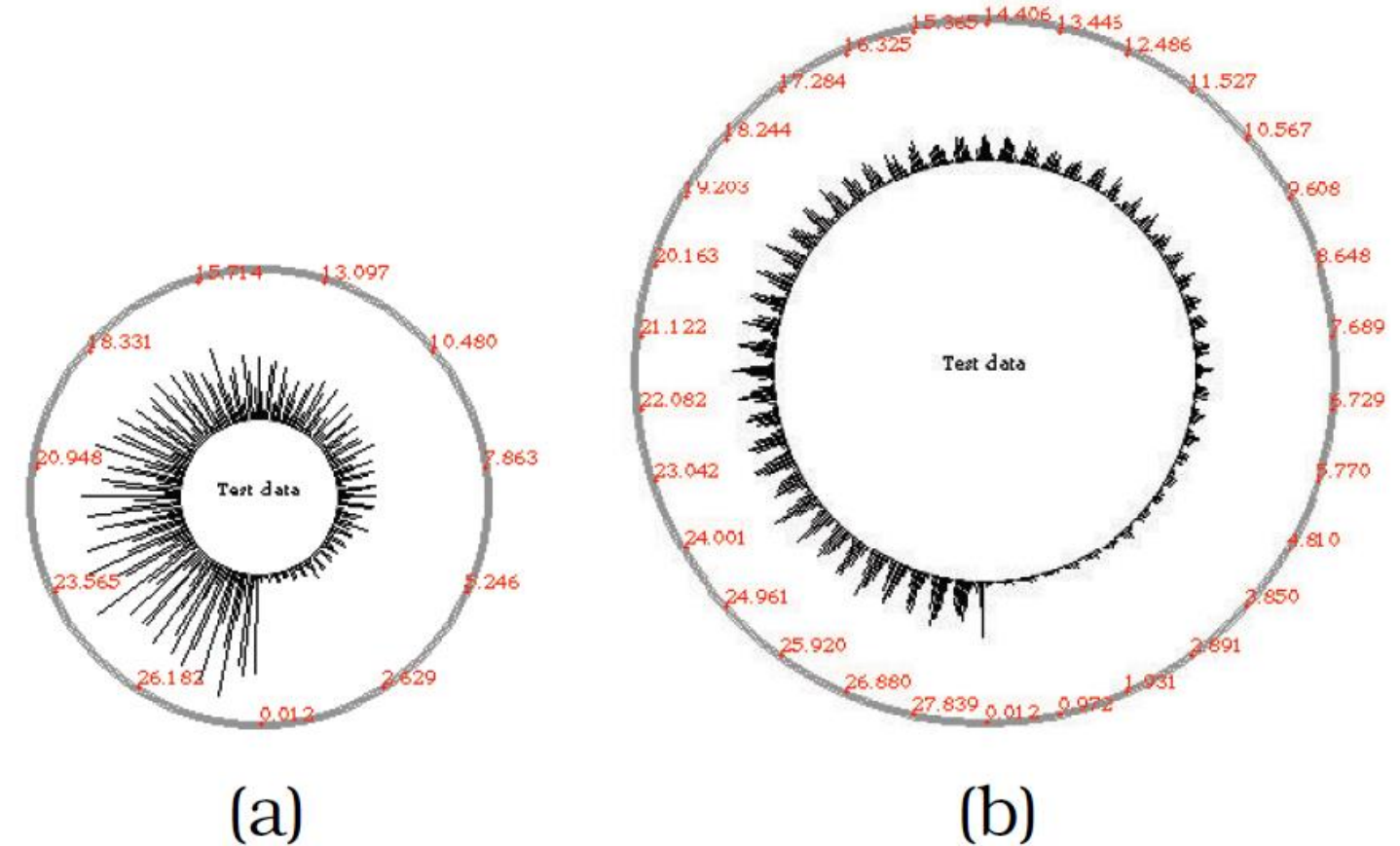
(a) Standard box plots - Presents summary statistics for the distribution of a quantitative attribute. These plots illustrate four kinds of distributions: normal (n), skewed (s), peaked (k), and multimodal (mm).

(b) Vase plots, which use horizontal spatial position to show density directly (good for mm).

From [Wickham and Stryjewski 12, Figure 5].

Idiom: Solar Plot, a radial histogram with an interactively controllable aggregation level [Chuah 98].

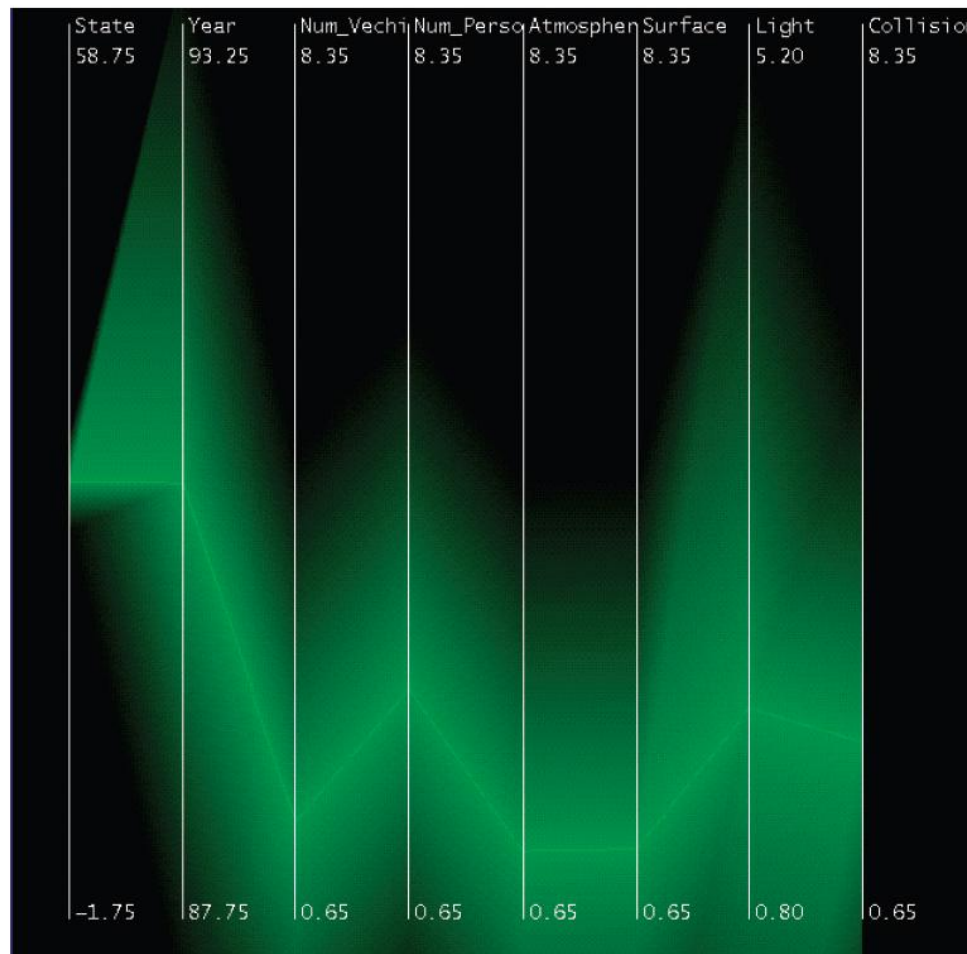
- The user directly manipulates the **size of the base circle** that is the radial axis of the chart. This change of radius indirectly changes the number of available histogram bins, and thus the aggregation level.



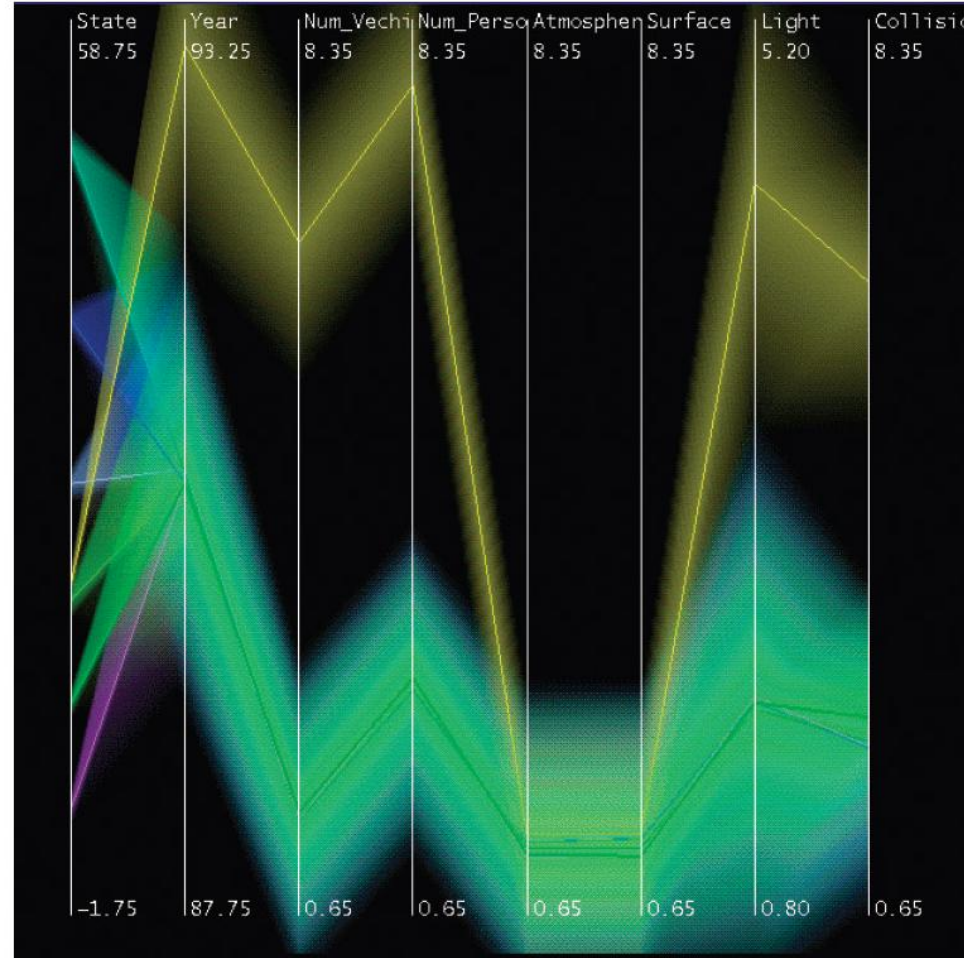
- (a) The small circle shows the increase in ticket sales over time (heavily aggregated)
- (b) Enlarging the circle shows seasonal patterns in addition to the gradual increase (less aggregated). From [Chuah 98].

- **Hierarchical aggregation** - constructs the derived data of a hierarchical clustering of items in the original dataset and allows the user to interactively control the level of detail to show.
- **Idiom: Hierarchical parallel coordinates** uses interactively controlled aggregation as a design choice to increase the scalability of the basic parallel coordinates visual encoding to hundreds of thousands of items.
- **Derived data:** a hierarchical clustering of the items, a cluster is represented by a band of varying width and opacity, where the mean is in the middle and width at each axis depends on the minimum and maximum item values for that attribute within the cluster.

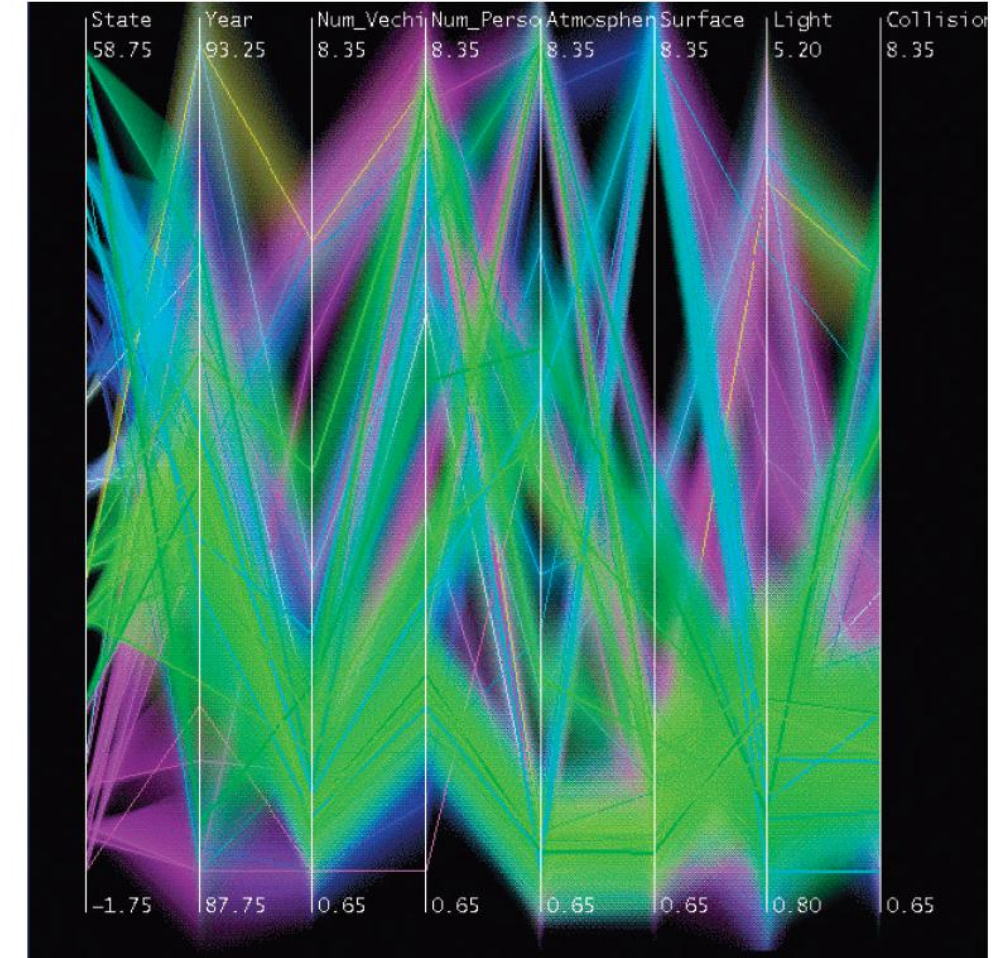
Item Aggregation



(a)



(b)



(c)

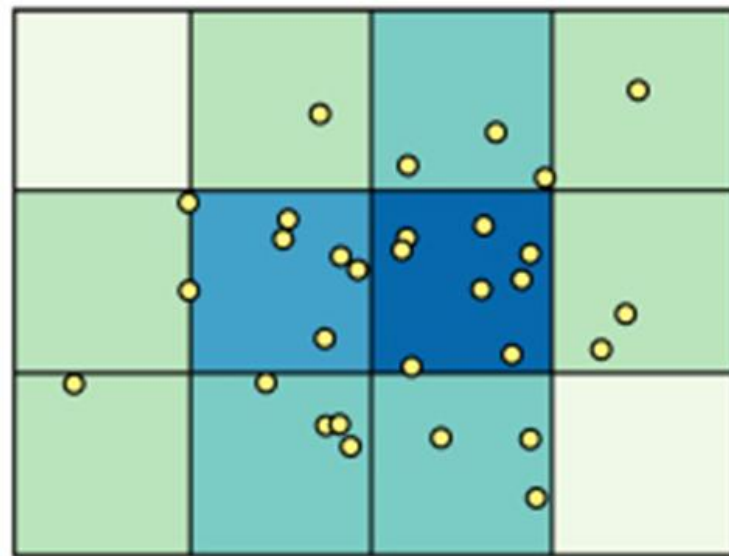
(a) The single top cluster has large extent.

(b) When several clusters are shown, each has a smaller extent.

(c) When many clusters are shown, the proximity-based colouring helps them remain distinguishable from each other. From [Fua et al. 99].

Spatial aggregation is to take the spatial nature of data into account correctly when aggregating it.

- In the cartography literature, the **modifiable areal unit problem (MAUP)** is a major concern: changing the boundaries of the regions used to analyse data can yield dramatically different results - gerrymandering (manipulating voting district boundaries) is one example!



Central region = high density



Medium density

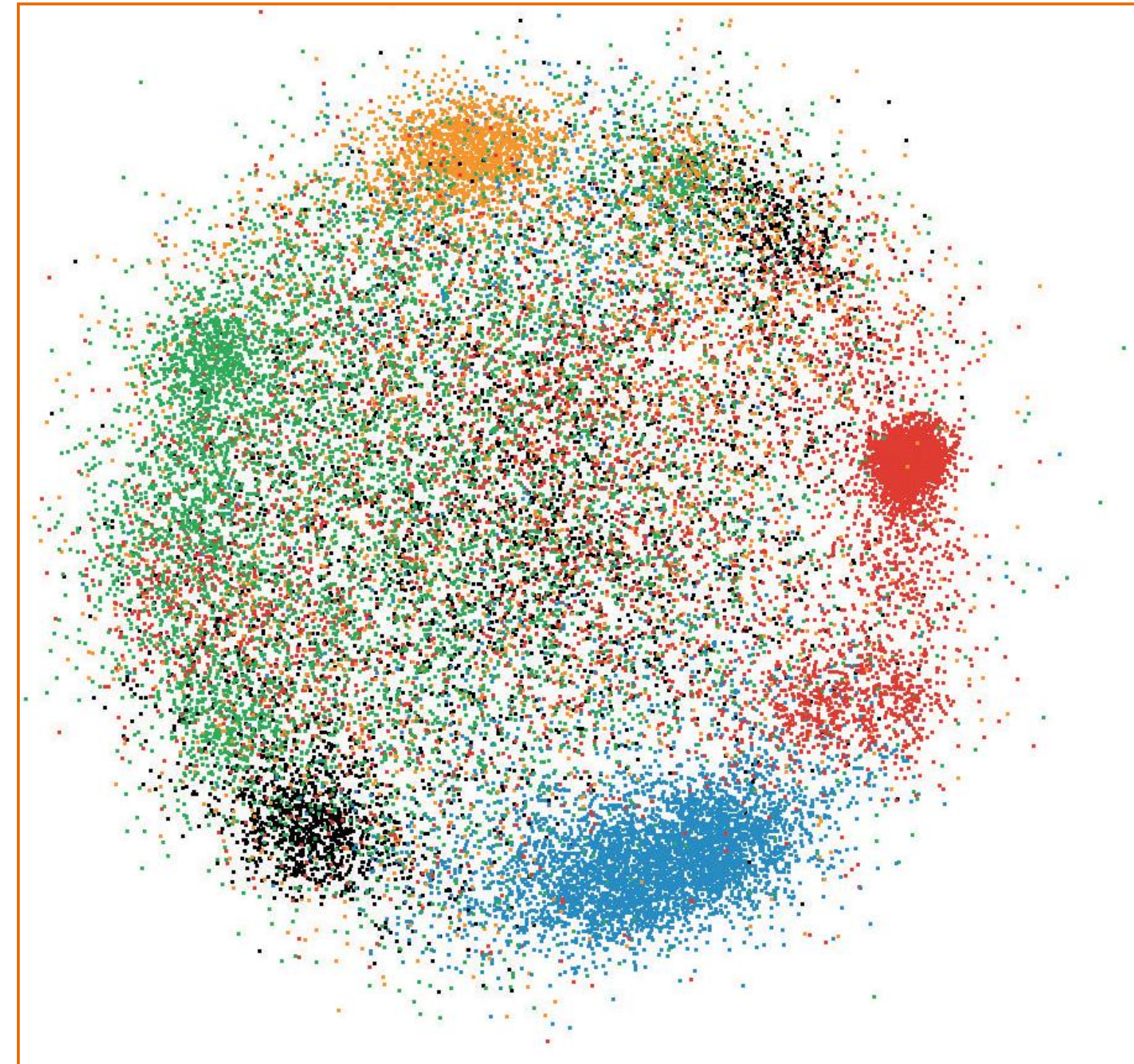


Low Density

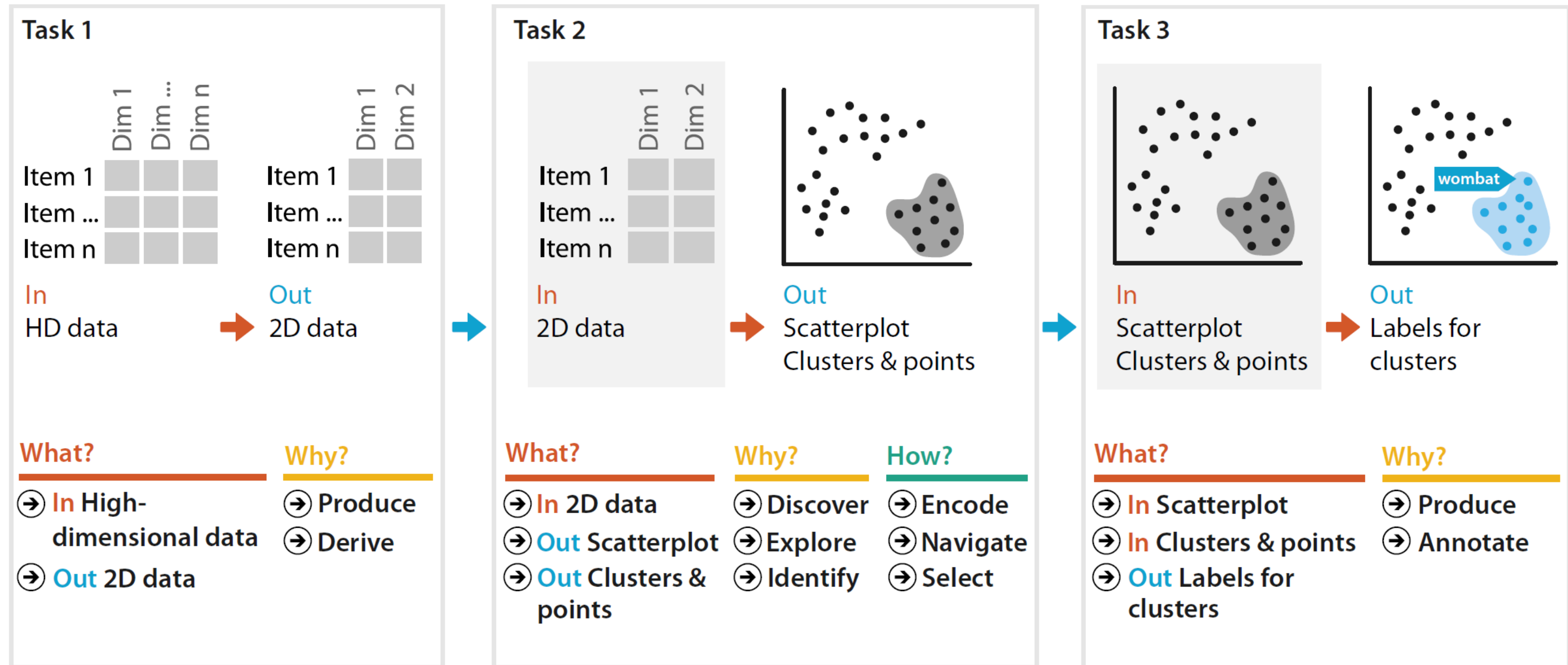
- Attributes can also be aggregated, where a new attribute is synthesised to take the place of multiple original attributes.
- A very simple approach to aggregating attributes is to group them by some kind of similarity measure, and then synthesise the new attribute by calculating an average across that similar set.
- A more complex approach to aggregation is **dimensionality reduction (DR)**, where the goal is to preserve the meaningful structure of a dataset while using fewer attributes to represent the items.

- Derive low-dimensional target space from high-dimensional measured space
- Use when you can't directly measure what you care about
- True dimensionality of dataset conjectured to be smaller than dimensionality of measurements

Dimensionality reduction of a large document collection (derived table with 10,000 attributes) using Glimmer for multidimensional scaling (MDS), -> a single 2D scatterplot (derived table with 2 attributes), allowing the user to verify that the conjectured clustering shown with colour coding is partially supported by the spatial layout. From [Ingram et al. 09, Figure 8].



Attribute Aggregation: Dimensionality Reduction



A **chained sequence** of what–why–how analysis instances for the scenario of dimensionality reduction of document collection data.